Class: 12 Physics (Marking Scheme and Micro-Syllabus)

Chapters	MCQs (1 mark)	SAQs (5 marks)	LAQs (8 marks)	Total marks
Mechanics	3	2	0	3+10=13
Thermodynamics	2	1	0	2+5=7
Wave & Optics	2	1	1	2+5+8=15
Electricity & Magnetism	3	2	1	3+10+8=21
Moderns Physics	1	2	1	1+10+8=19

1. Rotational Dynamics

1.1 Recall equations of angular motion and compare them with equations of linear motion
1.2 *Derive the expression for rotational kinetic energy*
1.3 Describe the term moment of inertia and radius of gyration
1.4 Find the moment of inertia of thin uniform rod rotating about its center and its one end
1.5 *Establish the relation between torque and angular acceleration of a rigid body*
1.6 Describe the work and power in rotational motion with expression
1.7 Define angular momentum and prove the principle of conservation of angular momentum

2. Simple Harmonic Motion

2.1 Define simple harmonic motion and state its equation.
2.2 ***Derive the expressions for energy in simple harmonic motion***
2.3 ***Derive the expression for period for vertical oscillation of a mass suspended from coiled spring (Hor.+ Vert.)***
2.4 Describe angular simple harmonic motion and find its period
2.5 ***Derive expression for period of simple pendulum***
2.6 Explain the damped oscillation
2.7 Describe forced oscillation and resonance with suitable examples

3. Fluid Statics

3.1 State and explain Archimedes principle and Pascal's law
3.2 Define up-thrust, pressure in fluid, buoyancy, center of buoyancy and meta center
3.3 State and use the law of floatation
3.4 Describe surface tension and explain its principle
3.5 ***Establish the relation between surface energy and surface tension***
3.6 Define angle of contact and capillarity with examples
3.7 State the Newton's Formula for viscosity of a liquid and define coefficient of viscosity
3.8 Recall and use the Poiseuilles' formula
3.9 Differentiate between laminar and turbulent flow & describe Reynolds number

3.10 Stokes law and its application
3.10 Equation of continuity and its application
3.11 Bernoulli's equation and its application

4. First law of thermodynamics

4.1 Clarify the concept of thermodynamic system. Explain the concept of potential energy, kinetic energy and internal energy.
4.2 Explain the meaning of work done by the system and work done on the system, and describe how work done by gas during expansion can be calculated from indicator (P – V) diagram.
4.3 State and explain first law of thermodynamics; realizing its limitations and necessity of second law of thermodynamics.
4.4 Define and explain two specific heat capacities of gas appreciating the relation $C_p - C_v = R$ and $c_p - c_v = r$.
4.5 Explain various thermodynamic process (isothermal, isobaric, isochoric and adiabatic) with good concept of their P – V diagram.
4.6 ***Derive adiabatic equation*** $\boldsymbol{PV^{\gamma} = constant}$.
4.7 ***Derive expression for work done during isothermal and adiabatic process.***
4.8 Give concept of reversible and irreversible process with examples.

5. Second Law of Thermodynamics

5.1 State and explain second law of thermodynamics (Kelvin's and Clausius's statement).
5.2 Compare second and first law of thermodynamics considering indication of direction of flow of heat.
5.3 Explain heat engine as a device to convert heat energy into mechanical energy appreciating that its efficiency is less than 100%.
5.4 Discuss Carnot's cycle with the concept of P – V diagram and calculate the work done of each step and corresponding efficiency.
5.5 Describe internal combustion engines, Otto engine and diesel engine with the help of P – V diagram to compare their efficiencies.
5.6 Explain refrigerator as heat engine working in reverse direction (coefficient of performance of refrigerator)
5.7 Introduce entropy as a measure of disorder appreciating its roles in thermodynamic process.

6. Wave motion

6.1 Introduction of wave and its types
6.2 Define and understand progressive wave
6.2 Write progressive wave in mathematical form
6.3 Discuss the condition under which stationary waves can be formed
6.4 Write stationary wave in mathematical form
6.5 Calculate frequency, amplitude, velocity, time period, etc. of progressive wave
6.6 ***Find expression for stationary wave using two progressive waves***

7. Mechanical Wave

7.1 Calculate Speed of wave motion (dimensional method)
7.2 Understand and write expression for the Velocity of sound in solid and liquid
7.3 ***Describe Velocity of sound in gas***
7.4 ***Describe Laplace correction***
7.5 ***Formulate the effect of temperature, pressure, humidity on velocity of sound and their physical meaning***

8. Wave in pipes and strings

8.1 Understand the formation of stationery waves in closed and open pipes
8.2 Define and understand harmonics and overtones
8.3 Discuss harmonics and overtones in closed and open organ pipes
8.4 Understand end correction in pipes
8.5 State and use the formula for velocity of transverse waves along a stretched string
8.6 Understand Vibration of string and overtones
8.7 Know the laws of vibration of fixed string.

9. Acoustic phenomena:

9.1 Describe sound waves as pressure waves in a medium (Obtain an expression for Pressure amplitude)
9.2 Obtain an expression for intensity of sound and show that intensity of sound obeys inverse square law
9.3 Characterize the sound using its intensity, loudness, quality and pitch
9.4 Discuss Doppler's effect
9.5 ***Apply Doppler effect in realistic case where source and observers are in relative motion.***

10. Nature and propagation of Light:

10.1 Nature of light- various models about nature of light
10.2 Wave-fronts, Wavelets , Types of wave-fronts
10.3 Use Huygens's principle to explain reflection and refraction of light

11. Interference

11.1 ***Explain the Phenomenon of Interferences*** Understand the meaning of coherent sources Describe the conditions for constructive and destructive interference
11.2 ***Describe Young's double slit experiment and obtain the expression for nth order maxima and minima***

11.3 Describe and apply the concept of optical path

12. Diffraction

12.1 Describe diffraction at a single slit
12.2 ***Understand diffraction pattern of image and derive the expression for the position of nth order minima***
12.3 Explain diffraction through transmission/diffraction grating and use the formula $d\ sin\theta\ =\ n\lambda$ for maxima
12.4 Explain resolving power of optical instruments

13. Polarization

13.1 Describe phenomenon of polarization
13.2 Explain how polarization of light explains the transverse nature of light
13.3 State and use Brewster's law

14. Electrical circuits

14.1 Understand Kirchhoff's law as well as use it to calculate unknown parameters in electrical circuits
14.2 Describe the circuit diagram and working of Wheatstone bridge circuit and understand its importance in real situation
14.3 Describe Meter bridge and understand it
14.4 Know construction, working and importance of Potentiometer
14.5***Show the understanding of construction, working principle and uses of Potentiometer for comparing emfs and measuring internal resistance of cells***
14.6 Understand the concept of super conductors
14.7 Know the meaning of perfect conductors and distinguish it from superconductor
14.8 Learn the technique to convert galvanometer into voltmeter and ammeter -Ohmmeter

15. Thermoelectric effects:

15.1 Explain See beck effect and its application in Thermocouples
15.2 Show understanding of the construction and working principle of thermocouple as a temperature measuring device
15.3 Explain Peltier effect
15.4 Understand the construction and working of Thermopile

16. Magnetic field:

16.1 Show understanding of the concept of magnetic field lines and magnetic flux and sketch magnetic field lines around a straight current carrying conductor and long solenoid
16.2 Explain Oersted's experiment, its outcome and limitations
16.3 Discuss force on moving charge in uniform magnetic field
16.4 Discuss force on a current carrying conductor placed in uniform magnetic field

16.5 Describe force and Torque on rectangular coil placed in uniform magnetic field
16.6 Describe moving coil galvanometer and know its applications
16.7 ***Explain Hall effect and derive the expression*** $V_H = \frac{BI}{nqt}$ ***where t is thickness***
16.8 Use Hall probe to measure flux density of a uniform magnetic field
16.9 State Biot and Savart law and know its application on (i) a circular coil (ii) a long straight conductor (iii) a long solenoid
16.10 State Ampere's law and know its applications to (i) a long straight conductor (ii) a straight solenoid (ii) a toroidal solenoid
16.11 Discuss force between two parallel conductors carrying current- definition of ampere

17. Magnetic Properties of materials:

17.1 Magnetic field lines and magnetic flus
17.2 Flux density in magnetic material: Relative permeability; Susceptibility
17.3 Diamagnetic, paramagnetic and ferromagnetic materials
17.4 Hysteresis

18. Electromagnetic Induction:

18.1 State and show understanding of Faraday's law of electromagnetic induction
18.2 State and show understanding of Lenz's law
18.3 Discuss construction and working of A.C. generators
18.4 Define eddy currents, explain how they arise and give a few examples where eddy currents are useful and where they are nuisance
18.5 Describe self-inductance and mutual inductance and understand their uses
18.6 State the expression for energy stored in an inductor and use it wherever needed
18.7 Discuss the construction, working principle and importance of transformer
18.8 Discuss the sources of energy loss in practical transformer

19. Alternating Currents:

19.1 Understand peak, mean and rms value of AC current and voltage
19.2 Discuss AC through a resistor, a capacitor and an inductor
19.3 Understand Phasor diagram in RC and RL circuits
19.4 Discuss series circuits containing combination of resistance, capacitance and inductance
19.5 Describe series resonance condition and know its applications
19.6 Understand the meaning of quality factor
19.7 Discuss power in AC circuits and know the term power factor

20. Electrons

20.1 Describe Millikan's oil drop experiment and explain how it suggests quantization of charge
20.2 ***Describe the motion of electrons in electric and magnetic fields and derive appropriate mathematical expressions***
20.3 ***Describe J.J Thomson's experiment with suitable diagrams to explain the discovery of electron and its characters, Cathode ray***

21. Photons

21.1 Describe quantum nature of radiation
21.2 Explain properties of photons
21.3 Describe work function and photoelectric effect
21.4 ***Derive Einstein's photoelectric equation***
21.5 Describe Millikan's experiment for the verification of Einstein's photoelectric equation and calculate Planck's constant

22. Semiconductor devices

22.1 Describe the formation of PN junction and semiconductor diode
22.2 Plot forward and reverse characteristics of semiconductor diode including the concept of Zener diode
22.3 Describe half and full wave rectification using semiconductor diodes Importance of filter circuit
22.4 Define logic gates and explain operation of different logic gates OR, AND, NOT, NAND and NOR gates with their symbol , Boolean algebra and truth table

23. Quantization of energy

23.1 Write the postulates of Bohr's model
23.2 ***Derive the expression of radius of nth orbit, velocity of electron in nth orbit and total energy of electron in nth orbit of H-atom***
23.3 ***Obtain the expression of wavelength of a spectral line***
23.4 ***Obtain mathematical expressions different spectral series of H-atom***
23.5 Differentiate excitation and ionization potentials
23.6 Explain emission and absorption spectra
23.7 Describe de Broglie hypothesis
23.8 Define x-rays Continuous and characteristics X-rays Illustrate different properties of x-rays/ Uses Soft and Hard X-rays
23.9 ***Describe modern Coolidge tube method for the production of x-rays with quality and quantity (intensity)***
23.10 Bragg's Law

24. Radioactivity and nuclear reaction

24.1 Explain the meaning of Radioactivity – natural and artificial Demonstration of radioactivity
24.2 Differentiate types of radiations coming from radioactive sources – alpha, beta particles and gamma rays and state their properties
24.3 Explain radioactive disintegration law ($\boldsymbol{N = N_o e^{-\lambda t}}$)
24.4 ***Obtain the expressions of half-life, decay constant and mean life***
24.5 ***Explain the working of Geiger-Muller Tube***
24.6 Analyze some medical uses and health hazard of nuclear radiation

25. Recent trends in physics

25.1 Seismology a. Briefly explain the origin of earthquakes b. Explain different types of surface waves: Rayleigh and Love waves c. Explain different types of internal waves: S and P-waves d. Give brief introduction to the wave patterns of Gorkha Earthquake 2015
25.2 Demonstrate basic ideas on a. Gravitational Wave b. Nanotechnology c. Higgs Bosons